

Chapter 16 Homework Answer Sheet

Part A Multiple choice (15 points)

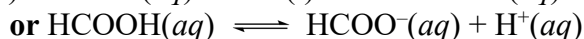
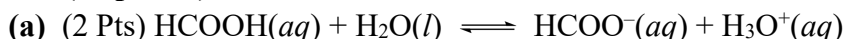
1-5 CBADB

6-10 ABECD

11-15 BEEED

Part B Short questions (15 points)

16. (12 points)



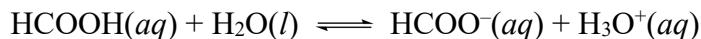
$$K_a = \frac{[\text{H}_3\text{O}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

or

$$K_a = \frac{[\text{H}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

(b) (3 Pts)

$$\text{p}K_a = 3.74 \quad K_a = 10^{-3.74} = 1.82 \times 10^{-4} = 1.8 \times 10^{-4}$$



Initial concentration / M	0.25	-	0	0
Change concentration / M	-x	-	+x	+x
Equilibrium concentration / M	0.25-x	-	x	x

$$\frac{x^2}{0.25 - x} = 1.82 \times 10^{-4}$$

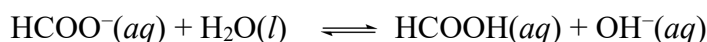
Assume $x \ll 0.25$, then $x^2/0.25 = 1.82 \times 10^{-4}$

$$[\text{H}_3\text{O}^+] = x = 0.00674 \text{ M}$$

$$\text{pH} = -\log[\text{H}_3\text{O}^+] = -\log(0.00674) = 2.17$$

(c) (4 Pts)

$$K_b = \frac{K_w}{K_a} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-4}} = 5.56 \times 10^{-11} = 5.6 \times 10^{-11}$$



I/M	0.20	--	0	0
C/M	-x	--	+x	+x
E/M	0.20-x	--	x	x

$$K_b = \frac{[\text{HCOOH}][\text{OH}^-]}{[\text{HCOO}^-]} = \frac{x^2}{0.20 - x} = \frac{x^2}{0.20} = 5.56 \times 10^{-11}$$

$$x = [\text{OH}^-] = 3.33 \times 10^{-6} \text{ M}$$

$$\text{pOH} = -\log[\text{OH}^-] = 5.478$$

$$\text{pH} = 14.00 - \text{pOH} = 14.00 - 5.478 = 8.52$$

(d) (3 Pts) The solution with 1 M NaH_2PO_4 is acidic.

K_b for hydrolysis of H_2PO_4^- is $K_w / K_{a1} = 1.0 \times 10^{-14} / (7.5 \times 10^{-3}) = 1.3 \times 10^{-12}$, which is smaller than K_{a2} (6.2×10^{-8}) for the dissociation of H_2PO_4^- .

17. (3 points)

(a) HY

(b) X^-

(c) The equilibrium will mostly lie to the left ($K_c < 1$).